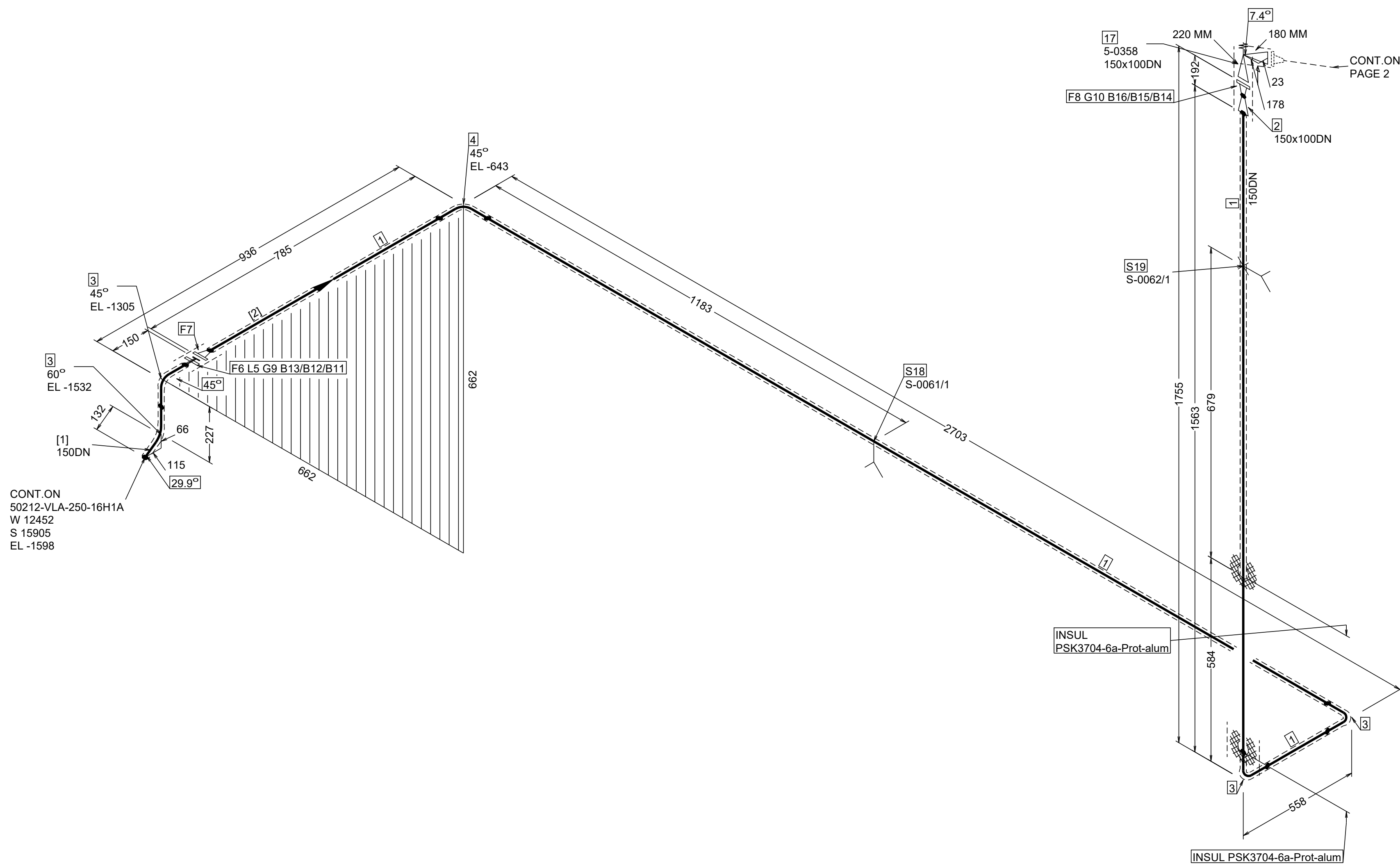


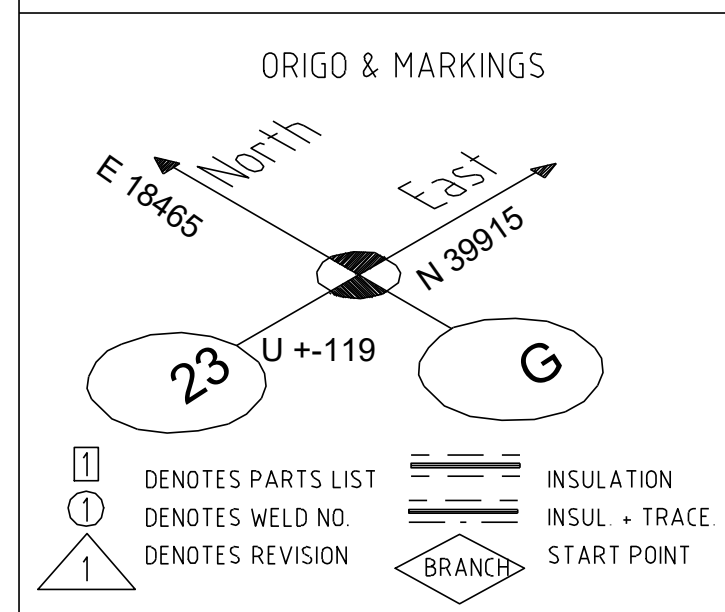
A
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F



NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 168.3 x 4.50 - EN 10216-2 - P235GH-TC1	150	119440-VE16C1C-R00	4448MM
2	Concentric reducer - EN 10253-2 - type A - 168.3 x 4.50 / 114.3 x 3.60 - P235GH-TC1	150 x 100	119440-VE16C1C-R00	1
3	Elbow - w - EN 10253-4 - type A - Table A.1 - model 3D - 90deg - 168.3 x 2.00 - 1.4307	150	119440-VE16H1A-R00	4
4	Elbow - EN 10253-2 - type A - model 3D - 90deg - 168.3 x 4.50 - P235GH-TC1	150	119440-VE16C1C-R00	1
5	Weldring neck - EN 1092-1 - type 35 - 168.3/DN150 - PN16 - 1.4307	150	119440-VE16H1A-R00	1
6	Loose flange - EN 1092-1 - type 02 - - 168.3/DN150 - PN16 - P265GH HDG	150	119440-VE16H1A-R00	1
7	Weld-neck flange - EN 1092-1 - type 11 - B1 - 168.3/DN150 - PN16 - P250GH	150	119440-VE16C1C-R00	1
8	Weld-neck flange - EN 1092-1 - type 11 - B1 - 114.3/DN100 - PN16 - P250GH	100	119440-VE16C1C-R00	1
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
9	Gasket - EN 1514-1 - type IBC - OD168.3/DN150 - 1.5mm Thk - PN16 Fiber	150	119440-VE16H1A-R00	1
10	Gasket - EN 1514-1 - type IBC - OD114.3/DN100 - 1.5mm Thk - PN16 Fiber	100	119440-VE16C1C-R00	1
11	90 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
12	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
13	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	20		16
14	70 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	16		8
15	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	16		8
16	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	16		16
17	KPK BASIC SAFETY VALVE FLANGED PN 16/16 5-0358	150 x 100	KBE/VALV01	1
	<u>SUPPORTS</u>			
18	HANGER PSK 7303 TYPE 12 See supportlist	150	VOFSUPP-PRIM-PSK-AT TA	1
19	2-SIDED ANCHOR PSK 7303 TYPE 10, LOW See supportlist	150	VOFSUPP-PRIM-PSK-AT TA	1

PIPE SPOOLS

[1] [2]



Bore	Thickness	Length	Insulation spec name
750	60mm	7644mm	/PSK3704-6a-Prot-alum
100	50mm	54mm	/PSK3704-6a-Prot-alum
150	50mm	284mm	/PSK3704-6a-Prot-alum
15	30mm	1277mm	/PSK3704-6a-Prot-alum

NOTE:

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.
Piping contractor is responsible for final routing and supporting

Welding factor	Acc. to pipe class
Efficiency of long weld	Acc. to pipe class
NDT circ. weld %	-
NDT long weld %	-
Welding list see doc.	-

		Hazardous CLP content		Degree of prefabrication to be decided by contractor.									
				Rev/ Revision Note		Created by		Checked by		Approved by		Date	
Directive	2014/68/EU	Pipe Class	/119440-VE16C1C-R00	0		J. Lakka		T. Saarinen		M. Kankkunen		5/5/2026	
Design press. (min)	-1bar	PED Category	I	Site Tervasaari, Valkeakoski		Equipment Type Höyry- ja lauhdejärjestelmä				Equipment ID			
Design temp. (min)	-	Media		Project Key TERPMS_NS265A		Project Name Ter PM 5 Safe Future				Project Number NS265A		Work	
Max. allow. press. (PS)	-	Standard		Projection Method		Welding Symbols ISO 5553		General Tolerances ISO 13 920-BF		ISO 2768-mK-E ISO 2768-mK-E		ISO 2768-mK-E ISO 2768-mK-E	
Operating press.	5bar	P&ID	RAU8F00303			This document is the exclusive intellectual property of Valmet Corporation and/or its subsidiaries (collectively and individually "Valmet") and is furnished solely for executing and maintaining the specific project. Re-use of the document for any other project or purpose is prohibited. The document or the information shall not be reproduced, copied or disclosed to any third party without the prior written consent of Valmet. By accessing this document, you accept these terms of use.							
Operating temp.	159degC	Design press	6bar										
PED Module	-	Design temp.	220degC	Title		50165-HMP-150-16C1B							
PED Group	-	Test press.	-										
Wind loads	-	Pipe list doc. ID	RAUAF04300	Status CERTIFIED		Document ID						Sheet 1 / 2	
External loads	-	Valve list doc. ID	RAUAF04301	Security Classification Internal								Revision 0	

A

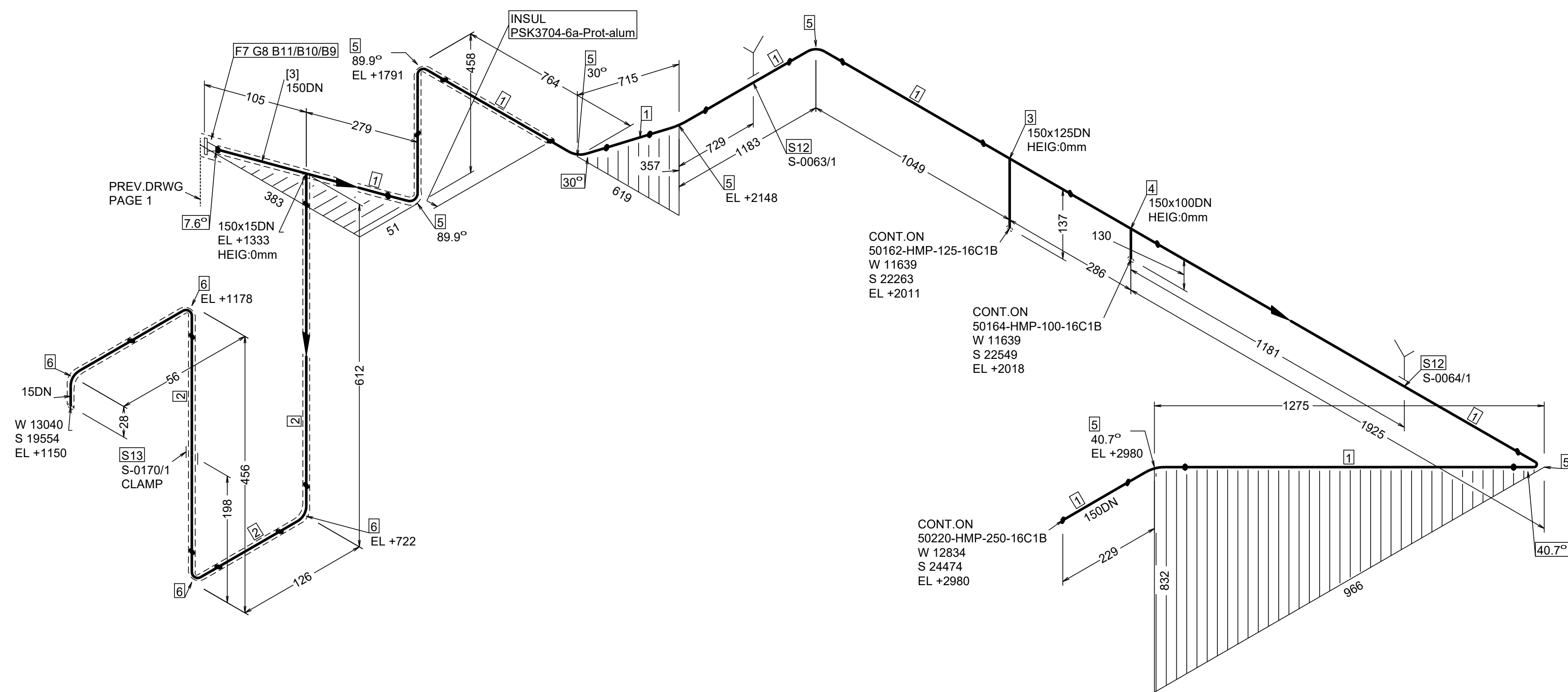
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E

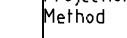
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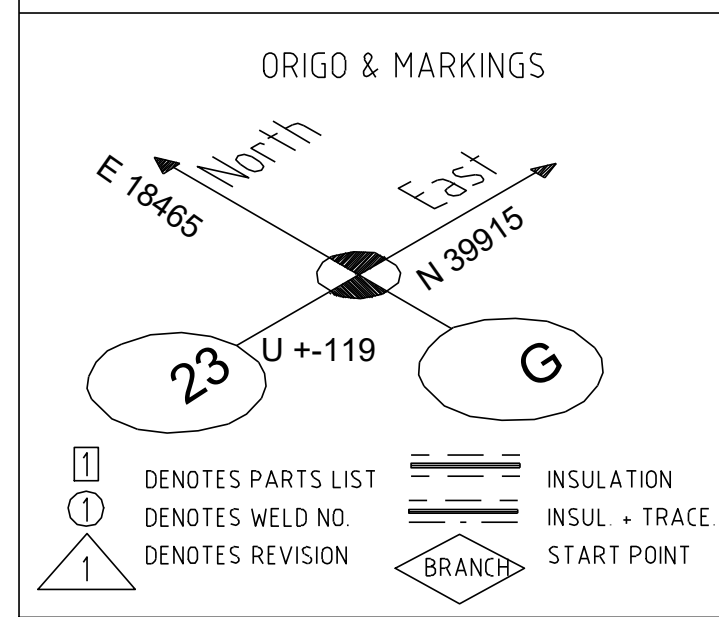
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
1	Tube - 168.3 x 4.50 - EN 10216-2 - P235GH-TC1	150	119440-VE16C1C-R00	5059MM
2	Tube - 21.3 x 2.00 - EN 10216-2 - P235GH-TC1	15	119440-VE16C1C-R00	970MM
3	Reduced tee - EN 10253-2 - type A - 168.3 x 5.60 / 139.7 x 4.00 - P235GH-TC1	150 x 125	119440-VE16C1C-R00	1
4	Reduced tee - EN 10253-2 - type A - 168.3 x 5.60 / 114.3 x 3.60 - P235GH-TC1	150 x 100	119440-VE16C1C-R00	1
5	Elbow - EN 10253-2 - type A - model 3D - 90deg - 168.3 x 4.50 - P235GH-TC1	150	119440-VE16C1C-R00	7
6	Elbow - EN 10253-2 - type A - model 3D - 90deg - 21.3 x 2.00 - P235GH-TC1	15	119440-VE16C1C-R00	4
7	Weld-neck flange - EN 1092-1 - type 11 - B1 - 168.3/DN150 - PN16 - P250GH	150	119440-VE16C1C-R00	1
NO	PART DESCRIPTION	DN (MM)	PIPE SPEC	QUANTITY
8	Gasket - EN 1514-1 - type IBC - OD168.3/DN150 - 1.5mm Thk - PN16 Fiber	150	119440-VE16C1C-R00	1
9	80 mm Bolt EN ISO 4014 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
10	Hexagon nut EN ISO 4032 25CrMo4, EN 10269, mat cert. 3.1, HDG	20		8
11	Washer EN ISO 7089 Grade A - S235JR EN10025-2 - HDG	20		16
	<u>SUPPORTS</u>			
12	SLIDE SUPPORT PSK 7303 TYPE 1, LOW See supportlist	150	VOFSUPP-PRIM-PSK-AT TA	2
13	SPECIAL SUPPORT See supportlist CLAMP	15	VOFSUPP-PRIM-PSK-AT TA	1

PIPE SPOOLS

[3]

		Hazardous CLP content		Degree of prefabrication to be decided by contractor.											
				Rev/ Revision Note		Created by		Checked by		Approved by		Date			
Directive		2014/68/EU		Pipe Class		/19440-VE16C1C-R00		J. Lakka		T. Saarinen		M. Kankkunen		5/5/2026	
Design press. (min)		-1bar		PED Category		I		Site Tervasaari, Valkeakoski		Equipment Type Höyry- ja lauhdejärjestelmä		Equipment ID			
Design temp. (min)		-		Media				Project Key TERPMS_N52654		Project Name Ter PM's Safe Future		Project Number NS265A		Work	
Max. allow. press. (PSI)		-		Standard				Projection Method  ISO 2553		General Tolerances ISO 2768-mK-E ISO 2768-mK-E ISO 13 920-BF ISO 2768-mK-E		Scale -		Size -	
Operating press.		5bar		P&ID		RAU8F00303				This document is the exclusive intellectual property of Valmet Corporation and/or its Subsidiaries (collectively and individually "Valmet") and is furnished solely for executing and maintaining the specific project. Re-use of the document for any other project or purpose is prohibited. The document or the information shall not be reproduced, copied or disclosed to any third party without the prior written consent of Valmet. By accessing this document, you accept these terms of use.					
Operating temp.		159degC		Design press.		6bar		Title 50165-HMP-150-16C1B							
PED Module		-		Design temp.		220degC									
PED Group		-		Test press.		-									
Wind loads		-		Pipe list doc. ID		RAUAF04300		Status CERTIFIED		Document ID					
External loads		-		Valve list doc. ID		RAUAF04301		Security Classification Internal				Revision 0			

For installation tolerances dimensions to be checked by pipe contractor at site.
Degree of prefabrication to be decided by contractor.



NOTE:

Site weldings to be added by contractor.

Buildings construction and equipment erection tolerances to be checked by contractor.

Pipe levels and dimensions are according to pipe center line.

Isometric drawings for sizes DN 15-25 are approximate.
Piping contractor is responsible for final routing and supporting.

Welding factor	Acc. to pipe class
Efficiency of long. weld	Acc. to pipe class
NDT circ. weld %	-
NDT long. weld %	-
Welding list see doc.	-